

Fayjullah Haque Emon

Location: Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

United International University

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GitHub Profile

LinkedIn Profile

EDUCATION

•United International University

B.Sc. in Computer Science and Engineering (CGPA: 3.59) | Major: Data Science

Feb. 2022 – Ongoing

Dhaka, Bangladesh

•Motijheel Boy's Govt. School & College

Higher Secondary Certificate (HSC) – Science (GPA: 5.00)

2018–2020

Dhaka, Bangladesh

•Sher-E-Bangla School & College

Secondary School Certificate (SSC) – Science (GPA: 5.00)

2016–2018

Dhaka, Bangladesh

RESEARCH & ACADEMIC WORK

•Microarchitectural Masking: A JIT-Based Adaptive Dispatcher for Resource-Resilient Serverless Computing [Dates]

Academic Research Paper

- Designed and evaluated Microarchitectural Masking, a zero-privilege, user-space JIT dispatcher that successfully neutralizes "Noisy Neighbor" interference in cloud hosts by dynamically shifting to hardware-pressure-optimized algorithmic kernels.
- Ensured virtually no performance loss for compute/IO workloads under contention while recovering ~60% of memory-bound latency.
- Mitigated an estimated 29.6kg of CO₂e per million invocations seamlessly without requiring underlying infrastructure changes.

•AI Driven Speech Based Depression Detection System [Dates]

Academic Research Paper

- Implemented machine learning and deep learning architectures including SVM, XGBoost, CNN, RNN, LSTM, and GMM models.
- Extracted robust vocal attributes utilizing openSMILE feature extraction tools and methodologies.
- Evaluated and analyzed system performance using Weighted Average Recall (WAR) and conducted comparative model analysis to optimize predictive reliability.

EXPERIENCE

•PHP Developer Intern [Dates]

Study.net

Dhaka, Bangladesh

- Gained hands-on experience in building and maintaining web applications in a professional environment using PHP.
- Collaborated with the development team to implement new features and optimize existing codebases.
- Focused on backend logic, database management, and ensuring cross-browser compatibility.
- Technologies used: PHP, MySQL, HTML/CSS, JavaScript.

PROJECTS

•Shelf To Tales [Dates]

Innovative 3D virtual bookshelf for book enthusiasts

- Created an interactive platform featuring a 3D visual bookshelf and personalized reading experiences.
- Integrated backend services to manage book catalogs and user reading history.
- Technologies: Java, Spring Boot, React.js (suggested), HTML/CSS.

•Smart Cooling Storage (Smart Farm Fresh Storage) [Dates]

IoT-driven automation system for post-harvest loss reduction

- Built an automated storage system featuring YOLOv8-based vegetable classification and robotic arm sorting.
- Implemented RFID-authenticated dashboards and IoT-driven temperature control via Peltier modules.
- Developed real-time monitoring features for weight sensing and storage conditions.
- Technologies: Python, YOLO, OpenCV, IoT, Arduino/ESP32, RFID.

•Software Testing Tool [Dates]

Comprehensive platform for API lifecycle management and test environment automation

- Supports creating multiple environments (Local, Staging) with custom variables utilizing the `{{variable_name}}` syntax framework.
- Automatically extracts and saves JWT authentication tokens to the active environment upon successful system login.
- Enables sending GET, POST, PUT, PATCH, and DELETE requests with custom headers and varied body structures.
- Structures API tests hierarchically into Workspaces, Modules, and timestamped Requests with definitive pass/fail feedback for high traceability.
- Supports exporting and importing the complete test suite as a single unified JSON file to streamline easy team collaboration.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, Java, C++, HTML, CSS

Frameworks & Tools: Spring Boot, Scikit-learn, OpenCV, YOLO, Figma, Postman, Canva

Web Technologies: PHP (Native/Laravel), MySQL, React.js, Bootstrap

Databases: MySQL, PostgreSQL

Software Quality Assurance: SQA Fundamentals, Automation Testing, Manual Testing

Expertise: Web Development, AI Research, IoT, Data Analysis, UI/UX Design

Soft Skills: Problem Solving, Team Collaboration, Analytical Thinking, Presentation

AWARDS & CERTIFICATIONS

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| • 5th Runner Up – UIU CSE Project Show - Microprocessor and Microcontroller Lab | <i>Summer 2025</i> |
| • Member – UIU Computer Club (General Member) | <i>Ongoing</i> |